

Chrisil Ouseph

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RESEARCH INTERESTS

My research lies at the interface of topology and the comparison theory of C^* -algebras. In particular, I have been trying to understand how much regularity properties such as pureness or $\mathcal{Z}$ -stability determine the homotopy behavior of Cuntz equivalence classes. I am also interested in the extent of the failure of regularity, measured by invariants like the radius of comparison or growth rank, arising due to various topological obstructions.

EDUCATION

Purdue University

Ph.D. in Mathematics; GPA: 4.0/4.0

Advisor: Andrew S. Toms.

Aug. 2022–Present

West Lafayette, IN

Indian Institute of Science Education and Research (IISER) Pune

B.S.–M.S. Dual Degree in Mathematics, with Distinction; CGPA: 9.2/10

Thesis Title: *Algebraic Topology and Obstruction Theory*.

Thesis Supervisor: Vivek M. Mallick.

Aug. 2017–May 2022

Pune, India

RESEARCH ARTICLES

- **Homotopies of Constant Cuntz Classes in Pure C^* -algebras.**

Joint with Andrew S. Toms. In preparation.

- **Contractible Cuntz classes in $\mathcal{Z}$ -stable C^* -algebras.**

Joint with Andrew S. Toms. Preprint, 2024. [arXiv:2412.12366](https://arxiv.org/abs/2412.12366).

SELECTED TALKS

- **Homotopies of Constant Cuntz Classes in Selfless C^* -algebras.** Great Plains Operator Theory Symposium (GPOTS), University of Iowa, Iowa City, IA. May 2026
- **Quantum Mechanics for the Working Mathematician.** Student Operator Algebras Seminar, Purdue University. Jan. 2026
- **Contractible Cuntz Classes in $\mathcal{Z}$ -stable C^* -algebras.** Great Plains Operator Theory Symposium (GPOTS), San Juan, Puerto Rico. Oct. 2025
- **Contractible Cuntz Classes in $\mathcal{Z}$ -stable C^* -algebras.** NSF/CBMS Regional Conference in the Mathematical Sciences: Classifying Amenable Operator Algebras, Texas Christian University, Fort Worth, Texas. Jun. 2025
- **The Modern Theory of Classification of C^* -algebras.** Student Operator Algebras Seminar, Purdue University. Apr. 2025
- **Lessons from My First Paper: Contractible Cuntz Classes Revisited.** Operator Algebras Seminar, Purdue University. Feb. 2025
- **Contractible Cuntz Classes.** Interactions between C^* -Algebraic KK-Theory and Homotopy Theory. Jan. 2025
- **Hilbert Hotels on Hilbert Spaces.** Student Operator Algebras Seminar, Purdue University. Sep. 2024
- **Formalizing Mathematics.** Student Colloquium, Purdue University. Sep. 2023

CONFERENCES AND WORKSHOPS

- Young Mathematicians in C^* -Algebras (YMC*A), University of Oxford, Oxford, United Kingdom. Aug. 2026
- Great Plains Operator Theory Symposium (GPOTS), University of Iowa, Iowa City, IA. May 2026
- Great Plains Operator Theory Symposium (GPOTS), San Juan, Puerto Rico. Oct. 2025
- NSF/CBMS Regional Conference in the Mathematical Sciences: Classifying Amenable Operator Algebras, Texas Christian University, Fort Worth, Texas. Jun. 2025
- Interactions between C^* -Algebraic KK-Theory and Homotopy Theory, online. Jan. 2025
- East Coast Operator Algebras Symposium (ECOAS), North Carolina State University, Raleigh, NC. Nov. 2024
- Noncommutativity in the North - MikaelFest, Chalmers University of Technology, University of Gothenburg, Sweden. Jun. 2024
- Oberwolfach Seminar: Classification of C^* -Algebras and Dynamics, MFO, Oberwolfach, Germany. May 2024
- East Coast Operator Algebras Symposium (ECOAS), Purdue University, West Lafayette, IN. Oct. 2023
- Groundwork for Operator Algebras Lecture Series (GOALS), Purdue University, West Lafayette, IN. Jun. 2023

RESEARCH SUPPORT AND VISITS

- Research Assistant, Department Summer Research Grant, Purdue University. Summer 2026
- Research Visit, University of Münster, Germany. May 2024

TEACHING

Purdue University, Indianapolis

- Recitation Instructor, MA 26200: Linear Algebra and Differential Equations. Spring 2026
- Recitation Instructor, MA 26200: Linear Algebra and Differential Equations. Fall 2025

Purdue University, West Lafayette

- Instructor, MA 59800: Graduate Preparatory Course for the Real Analysis Qualifying Exam. Summer 2025
- Teaching Assistant, MA 44200: Honors Real Analysis II. Spring 2025
- Instructor, MA 16010: Applied Calculus I. Fall 2024
- Course Coordinator, MA 16200: Plane Analytic Geometry and Calculus II. Summer 2024
- Recitation Instructor, MA 262: Linear Algebra and Differential Equations. Spring 2024
- Recitation Instructor, MA 261: Multivariate Calculus. Fall 2023
- Teaching Assistant, MA 303: Ordinary and Partial Differential Equations. Summer 2023
- Teaching Assistant, MA 366: Ordinary Differential Equations. Spring 2023
- Recitation Instructor, MA 161: Plane Analytic Geometry and Calculus I. Fall 2022

ACADEMIC SERVICE

- **Co-organizer, AMS Special Session** on Recent Advances in Operator Algebras, Chicago, IL. Jan 2027
- **Direct Reading Program, Purdue University.** Supervised a semester-long undergraduate reading project entitled *Nets: Convergence in General Topological Spaces*. Fall 2025
- **Co-organizer, Student Operator Algebras Seminar**, Purdue University. Aug. 2024–Present